

ASH Sector-Mixing Resolution: Implementation Pass 002

Independent proposed extension; the original ASH repository is used only as a reference.

Canonical context

The ASH reference describes a 256-state parity-valid physical state space inside F_2^9 , with x_9 recomputed from the first eight bits, and a baseline lazy pair-flip kernel. The repository also labels physical interpretations and external validation as open research gates.

Blocker handled

The pair-flip kernel preserves Hamming-weight parity. It therefore splits the admissible state space into two closed 128-state components. This package implements a mathematically minimal sector-refresh term.

Resolution

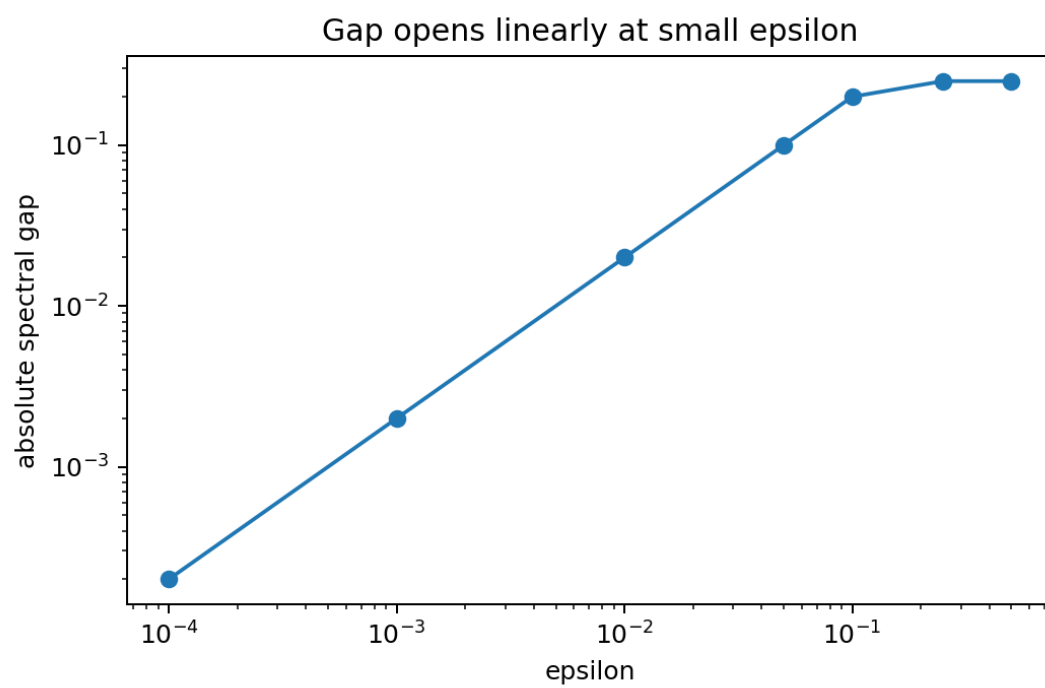
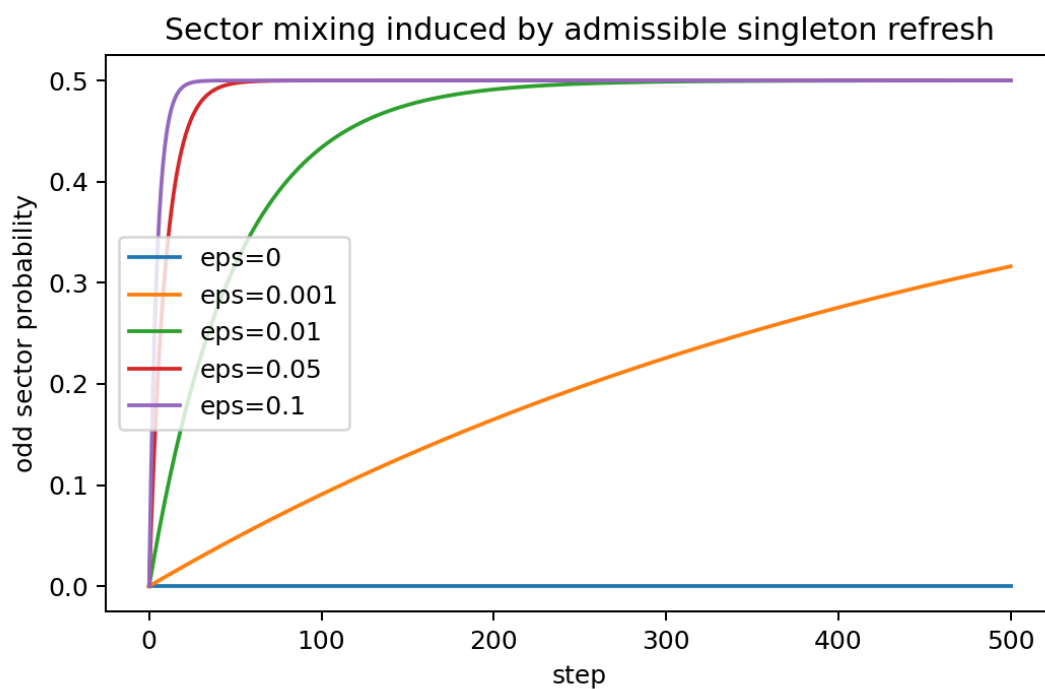
Use $K_{\epsilon} = (1-\epsilon) K_{\text{pair_lazy}} + \epsilon S$, where S flips one of the eight free bits uniformly and recomputes x_9 . This preserves admissibility but toggles the sector, making the chain connected.

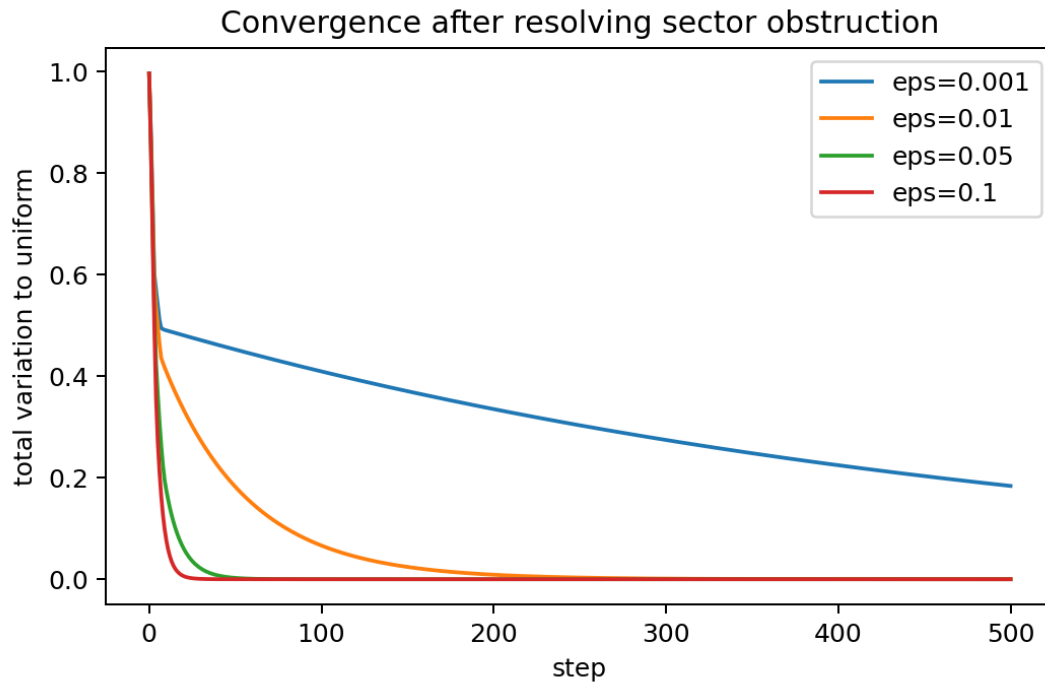
Result

For $0 < \epsilon < 1$ the kernel is symmetric, stochastic, connected, and aperiodic; uniform measure on the 256 admissible states is the unique stationary law. The former sector eigenmode becomes $1 - 2\epsilon$.

Spectral scan

epsilon	connected	gap	lambda2	max_formula_error
0.0	False	0	1	1.44e-15
0.0001	True	0.0002	0.9998	1.33e-15
0.001	True	0.002	0.998	1.44e-15
0.01	True	0.02	0.98	1.78e-15
0.05	True	0.1	0.9	1.33e-15
0.1	True	0.2	0.8	1.33e-15
0.25	True	0.25	0.75	1.05e-15
0.5	True	0.25	0.75	1.33e-15





Falsifiability and next steps

The repair creates a tunable prediction: sector imbalance decays approximately with eigenfactor 1-2 epsilon. Any future bridge map or empirical calibration that requires persistent sector separation must set epsilon to zero and reinterpret the sectors as superselection. Otherwise epsilon must be calibrated or bounded by external data before physical claims are made.